In [3]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Load dataset
data = pd.read_csv('../dataSet/study_exam_data.csv')
df = pd.DataFrame(data)
```

In [4]:
```python
# Fit linear regression using scipy.stats.linregress
slope, intercept, r_value, p_value, std_err = stats.linregress(df['Study_Hours'], df['Exam_Score'])

# Print regression equation, R^2, and p-value
print(f"Regression equation: Exam_Score = {intercept:.2f} + {slope:.2f} * Study_Hours")
print(f"R-squared: {r_value**2:.2f}")
print(f"P-value for slope: {p_value:.4f}")

# Predict exam score for 10 study hours
pred_score_10 = intercept + slope * 10
print(f"Predicted exam score for 10 study hours: {pred_score_10:.2f}")
```

```
Regression equation: Exam_Score = 52.19 + 2.83 * Study_Hours
R-squared: 0.59
P-value for slope: 0.0000
Predicted exam score for 10 study hours: 80.53
```

In [7]:
```python
# Plot data points and regression line
sns.scatterplot(x='Study_Hours', y='Exam_Score', data=df)
x = np.array(df['Study_Hours'])
y = intercept + slope * x
plt.plot(x, y, color='red')
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.grid(True)
plt.title('Study Hours vs Exam Score')
plt.show()
```